

# Predicting Contract Renewal in Public Digital Procurement

A Survival-Analysis Approach on BOAMP Notices for Gigalis  
Internship Report — Executive Orientation

Internship Report — Gigalis (Pays de la Loire)

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## Abstract

This is the **executive report** of the internship: it lets a reader scan the project rapidly and understand *what was done*, *why it matters*, and *what is reliable*. The full implementation and diagnostics are in the **technical report** (`phase1_technical_report.pdf`); the data reliability and event-definition detail are in the **data quality report**.

The work concerns predictive analysis of public digital procurement for **Gigalis**, a central purchasing body in Pays de la Loire. The goal is to move from a descriptive view (“what happened?”) towards a predictive one (“when will a contract be re-tendered?”). Because BOAMP contains *no explicit renewal field*, the renewal relationship is reconstructed from observable signals: a proxy renewal **event** built from Sentence-Transformer text embeddings combined with CPV, temporal, and buyer signals, used as the event variable for survival analysis. On 1,210 eligible source contracts the linking method recovers a plausible renewal for 665 of them (**55.0%**), against a 7.6% lexical baseline at the same window setting. Survival modeling gives a global Kaplan–Meier median survival of **50.1 months**, a Cox concordance of **0.6317**, and a best-fitting log-normal parametric model (AIC 7,068.4). The fitted model is translated into 12- and 24-month renewal-risk indicators at contract, buyer, and segment level. Buyer identities were consolidated by a SIREN enrichment step (completed 2026-06-24). The headline caveat: **event** is a proxy, and a manual audit of a stratified 150-case sample (2026-07-02) estimates its precision at only  $\approx 9\text{--}15\%$  overall — concentrated in the high-text-similarity core (precision 1.00 at text  $\geq 0.80$ ,  $\approx 0.50$  in the HIGH/strict tier,  $\approx 0$  below composite 0.50). The survival results should be read as describing algorithm-identifiable same-buyer re-publications, not validated renewals.

## 1 Context and Objective

### 1.1 Why this project

Gigalis operates a central purchasing body for digital services in Pays de la Loire. To anticipate re-tendering workloads, it wants to predict *when* digital contracts come up for renewal. The core obstacle: BOAMP publishes calls for tenders and awards, but *no notice says “this renews contract X”*. The renewal link must be inferred from text, classification, timing, and buyer identity.

### 1.2 Objective

The internship objective is to build a transparent, reproducible pipeline that (i) reconstructs a proxy renewal **event** from BOAMP signals, (ii) uses it as the outcome of a survival model of contract lifetime, and (iii) translates the model into operational renewal-risk indicators. This report orients

the reader to that pipeline and its main results; exact field logic, formulas, and diagnostics are deferred to the technical and data quality reports.

## 2 Data Sources and Scope

### 2.1 Scope

- **Sectors:** CPV divisions 48 (software), 72 (IT services), 32 (telecom), 35 (security).
- **Territory:** departments 44, 49, 53, 72, 85 (Pays de la Loire).
- **Period:** 2015–2024. Study end date: 2024-12-31.
- **Modeling path:** BOAMP only. DECP is documented but not linked (no shared key between the two sources).

### 2.2 Corpus numbers

- 3,181 BOAMP notices downloaded (2015–2024).
- 1,933 are APPEL\_OFFRE (source contracts); 1,081 are ATTRIBUTION.
- 1,210 are *eligible*: their expected renewal window ( $\pm 6$  months around the estimated end date) closes before 2024-12-31.
- 723 are structurally censored upfront (too recent to observe a renewal).

DECP (3,039 digital contracts, same scope) is far richer (100% SIRET, 99.4% durations) but cannot be merged to BOAMP at contract level; contract-level DECP merging remains a future path.

### 2.3 Key data quality issues (modeling consequences)

**Table 1:** Key BOAMP data-quality findings and their modeling consequence. Full field-level detail is in the data quality report.

| Finding                        | Value                  | Consequence                                                                          |
|--------------------------------|------------------------|--------------------------------------------------------------------------------------|
| Buyer SIRET coverage (raw)     | ~9% of corpus          | Grouping uses normalised names; SIREN enrichment later consolidates many (see §3.2). |
| Amount fill rate (APPEL_OFFRE) | ~21%                   | Amount excluded from linking signals.                                                |
| Declared duration granularity  | 69% at 12/24/36/48 mo  | Duration used as a covariate, not as survival target.                                |
| Imputed durations              | 25% of eligible cohort | Flagged with <code>dur_was_imputed</code> ; imputation is 48 months.                 |
| No explicit renewal label      | structural             | <code>event</code> is a proxy reconstructed by the linking algorithm.                |

### 2.4 Taxonomy

Each contract is tagged with one of eleven technology categories (Software & Applications, IT Services & Consulting, Telecom & Networks, Cybersecurity, Cloud & Infrastructure, Data & AI, etc.) via a **deterministic rule-based mapping** from CPV codes and description keywords. This is *not* a trained classifier — the planned NLP classifier was not implemented. Output: `data/processed/taxonomy.csv`.

### 3 Main Pipeline

The pipeline runs end to end from raw BOAMP notices to operational risk indicators. Table 2 summarises each component and where its outputs live; the subsections below describe what each stage does, and Section 4 reports the numbers.

**Table 2:** Component status and key outputs. Status: **Done**, **Partial**, **Not started**.

| Component                              | Status   | Key result                                               | Where                                              |
|----------------------------------------|----------|----------------------------------------------------------|----------------------------------------------------|
| Data cleaning & EDA                    | Done     | 3,181 notices → 1,933 source contracts                   | data/processed/boamp_full_clean.csv                |
| Buyer SIREN enrichment                 | Done     | 724/1,210 eligible rows on a SIREN key                   | buyer_siren_enrichment/                            |
| Renewal linking (event reconstruction) | Done     | 665/1,210 eligible linked (55.0%)                        | boamp_renewal_linking_quality/                     |
| Survival dataset construction          | Done     | 1,210 rows × 33 cols                                     | data/processed/boamp_phase2_survival.csv           |
| Kaplan–Meier & log-rank                | Done     | Median 50.1 mo; duration significant                     | notebooks/02_survival_modeling_boamp.ipynb         |
| Cox PH model                           | Done     | C-index 0.6317                                           | same notebook + reports/tables/survival/           |
| Parametric AFT models                  | Done     | Log-normal best (AIC 7,068.4)                            | same notebook                                      |
| Sensitivity analysis                   | Done     | Directions stable (A/B); strict variant underpowered     | reports/tables/survival/sensitivity_comparison.csv |
| Robustness checks                      | Done     | Placebo, bias check, threshold sweep                     | validation_robustness/                             |
| Operational risk indicators            | Done     | 12/24-mo risk at contract/buyer/segment                  | reports/tables/survival/top20_renewal_risk.csv     |
| Manual event validation                | Done     | 150 cases audited; precision ≈0.09–0.15 (HIGH tier 0.50) | event_validation/manual_validation_summary.md      |
| Trained NLP classifier                 | Not done | Taxonomy is rule-based only                              | —                                                  |
| Change-point detection (Phase 4)       | Not done | Scoped but not implemented                               | —                                                  |

### 3.1 BOAMP cleaning

Raw notices are parsed from the BOAMP API, deduplicated, and standardised: buyer key construction, contract start-date refinement (using `ATtribution` back-references where available), duration cleaning and imputation (missing eligible durations set to 48 months, flagged), CPV normalisation, and text normalisation. Output: `data/processed/boamp_full_clean.csv`.

### 3.2 Buyer SIREN enrichment

Because raw SIRET coverage is weak ( $\sim 9\%$  of the corpus), buyers were initially grouped by normalised name, which fragments a single public entity across spelling variants. A **SIREN enrichment step** (completed **2026-06-24**) queries the official *API Recherche d'Entreprises* for each buyer name and, for **HIGH-confidence matches only**, upgrades the buyer key from `NAME:` to `SIREN:{9-digit}`, merging name aliases of the same legal entity. Of the 1,210 eligible contracts, **724 carry a SIREN key (plus 2 SIRET) and 484 remain name-keyed**; 275 of 525 unique buyers (52.4%) matched with HIGH confidence. Medium/low matches are kept for review but never applied automatically. Enrichment is a buyer-resolution improvement, *not* ground truth: rare false merges at HIGH confidence remain possible. Outputs: `buyer_siren_enrichment/` and `data/processed/boamp_phase2_survival_siren_enriched.csv`.

### 3.3 Renewal-event reconstruction

Each eligible source contract is matched against later BOAMP notices from the same buyer using a four-signal composite score: semantic text embeddings (Sentence-Transformer, `paraphrase-multilingual-MiniLM-L12-v2`), CPV hierarchy match, temporal proximity, and buyer-identity agreement (weights 0.40 / 0.25 / 0.20 / 0.15). The best candidate passing the hard text-similarity filter is selected as the renewal successor and sets `event = 1`; contracts with no qualifying successor before study end are right-censored (`event = 0`). See technical report §5 for the full formula.

### 3.4 Survival modeling

The survival dataset (`data/processed/boamp_phase2_survival.csv`) has one row per eligible contract with observed/censored duration and covariates (`declared_duration_months`, `dur_was_imputed`, `start_year`, category, procedure type). Three model families were fitted in `notebooks/02_survival_modeling_boamp.ipynb`: Kaplan–Meier (with log-rank tests), a Cox proportional-hazards model, and parametric AFT models (Weibull, log-logistic, log-normal). The fitted model was stress-tested with a sensitivity analysis over three event definitions and a battery of robustness checks.

### 3.5 Operational risk indicators

The best parametric model (log-normal AFT) is applied to all scored contracts to produce conditional renewal probabilities at 12 and 24 months ahead, aggregated to contract, buyer, and segment level. These are prioritisation indicators, not legally verified renewal probabilities. Output: `reports/tables/survival/top20_renewal_risk.csv`.

## 4 Main Results

### 4.1 Renewal linking

**Table 3:** Renewal-linking outcome.

| Quantity                                      | Value |         |
|-----------------------------------------------|-------|---------|
| Source contracts ( <code>APPEL_OFFRE</code> ) | 1,933 |         |
| Eligible (renewal window observable)          | 1,210 |         |
| Linked renewals ( <code>event</code> = 1)     | 665   | (55.0%) |
| Right-censored ( <code>event</code> = 0)      | 545   | (45.0%) |
| Lexical baseline linking rate ( $W = 6$ )     | 7.6%  |         |

Linking rate by category (Table 4): Software & Applications and IT Services & Consulting renew most (62.8%, 60.5%); Data & AI and IT Hardware & Equipment least (30.0%, 34.6%). Small categories ( $n < 10$ ) are underpowered.

**Table 4:** Event rate by technology category (1,210 eligible contracts; events sum to 665).

| Category                          | n   | events | rate % |
|-----------------------------------|-----|--------|--------|
| IT Services & Consulting          | 344 | 208    | 60.5   |
| Software & Applications           | 223 | 140    | 62.8   |
| Telecom & Networks                | 223 | 116    | 52.0   |
| Cybersecurity                     | 136 | 73     | 53.7   |
| Unknown                           | 115 | 49     | 42.6   |
| Digital Workplace & Collaboration | 51  | 32     | 62.7   |
| Data & AI                         | 40  | 12     | 30.0   |
| IT Maintenance & Support          | 22  | 11     | 50.0   |
| IT Hardware & Equipment           | 26  | 9      | 34.6   |
| Cloud & Infrastructure            | 22  | 11     | 50.0   |
| GIS & Mapping                     | 8   | 4      | 50.0   |

**Link quality.** Of the 665 accepted links: 164 are single-candidate (no competing match); of the remaining 501 multi-candidate links, 219 (44%) have a score margin below 0.05, flagging an ambiguous winner. 75 links (11.3%) satisfy the `high_confidence_strict` criterion ( $S \geq 0.70$  and margin  $\geq 0.05$ , multi-candidate only); these form the Model C core (Table 7). **Main unlinked failure mode:** 491 of 545 unlinked contracts (90.1%) have no temporal partner in the search window — a true absence of a candidate notice, not a scoring failure.

### 4.2 Survival analysis

**Kaplan–Meier.** Global median survival is **50.1 months**, consistent with the prevalence of 48-month framework contracts. Log-rank tests (Table 5): renewal timing is driven by declared duration and imputation status ( $p < 0.001$ ); category is marginally significant ( $p = 0.025$ ).

**Table 5:** Log-rank tests for differences in renewal timing.

| Stratifying variable       | statistic | p       | df |
|----------------------------|-----------|---------|----|
| Category (multivariate)    | 12.89     | 0.025   | 5  |
| Duration was imputed       | 19.96     | < 0.001 | 1  |
| Declared-duration group    | 41.05     | < 0.001 | 2  |
| Procedure type (collapsed) | 3.48      | 0.176   | 2  |

**Cox proportional-hazards model.** In-sample concordance: **0.6317**. Two predictors are significant (Table 6): longer declared duration lowers hazard (HR 0.991 per month), and later start year raises it (HR 1.071 per year). IT Services & Consulting shows a positive category effect (HR 1.230,  $p = 0.037$ ). The positive `start_year` hazard is partly structural: 2021–2024 contracts are right-censored because a typical 48-month contract would renew after the 2024-12-31 study close.

**Table 6:** Cox model: significant covariates (multivariate).

| Covariate                                    | HR                            | 95% CI         | p       |
|----------------------------------------------|-------------------------------|----------------|---------|
| <code>declared_duration_months</code>        | 0.991                         | [0.986, 0.995] | < 0.001 |
| <code>start_year</code>                      | 1.071                         | [1.031, 1.113] | < 0.001 |
| IT Services & Consulting (vs. Cybersecurity) | 1.230                         | [1.013, 1.493] | 0.037   |
| <code>dur_was_imputed</code>                 | 0.871                         | [0.722, 1.049] | 0.146   |
| Other category / procedure terms             | not significant ( $p > 0.1$ ) |                |         |

**Parametric AFT models.** Log-normal fits best (AIC 7,068.4, concordance 0.655), outperforming log-logistic (7,154.7) and Weibull (7,240.3). Full results: `reports/tables/survival/parametric_aic_comparison.csv`.

**Sensitivity to event definition.** Three event definitions were tested (Table 7). The Cox concordance drops from 0.632 (Model A) to 0.616 (Model B,  $\geq 0.50$ ) and 0.455 (Model C, strict H.C.), suggesting that low-confidence links add predictive signal not present in the strict subset — and that the 75-event strict definition is too sparse for stable estimation.

**Table 7:** Sensitivity to event definition. “—” = KM median never reached. KM median in months.

| Variant                                                             | events | rate % | KM median | Cox C |
|---------------------------------------------------------------------|--------|--------|-----------|-------|
| A — baseline (all links)                                            | 665    | 55.0   | 50.1      | 0.632 |
| B — conservative ( $\geq 0.50$ )                                    | 400    | 33.1   | —         | 0.616 |
| C — strict H.C. ( $S \geq 0.70$ , $\Delta \geq 0.05$ , multi-cand.) | 75     | 6.2    | —         | 0.455 |

**Proportional-hazards assumption.** The Schoenfeld test flags a violation for `declared_duration_months` ( $p < 10^{-5}$ ); all other covariates pass. Stratification or a time interaction for duration is future work. The log-normal AFT avoids the PH assumption and gives the same conclusions.

### 4.3 Operational risk indicators

**Table 8:** Operational risk summary (1,204 scored contracts).

| Metric                                      | Value                           |
|---------------------------------------------|---------------------------------|
| Mean 12-month renewal probability           | 0.074                           |
| Mean 24-month renewal probability           | 0.216                           |
| Max 12-month probability (any contract)     | 0.288                           |
| Contracts in <i>medium</i> band (0.25–0.50) | 6                               |
| Contracts in <i>low</i> band ( $< 0.25$ )   | 1,198                           |
| Expected renewals within 12 months          | 89.7                            |
| Expected renewals within 24 months          | 259.9                           |
| Top buyer by expected 12-month volume       | SIREN:234400034 (9.3)           |
| Top segment by expected 12-month volume     | IT Services & Consulting (30.4) |

## 5 Reliability and Limitations

### 5.1 Reliability evidence

The event variable was stress-tested with checks in `validation_robustness/`:

- **Threshold sensitivity (7 scenarios: baseline, composite floors 0.50–0.90, strict H.C.):** Cox concordance declines gradually from 0.632 (baseline) to 0.599 (score  $\geq 0.60$ ), then becomes unstable below  $\approx 100$  events (0.473 at  $\geq 0.70$ ); hazard-ratio directions stay stable at every threshold with enough events. Full table: `validation_robustness/outputs/threshold_sensitivity_summary.csv`.
- **Placebo check:** winning-link mean score 0.554 vs. runner-up 0.428 vs. synthetic no-buyer 0.401 — the scoring discriminates (14.9% of winners reach  $\geq 0.70$  vs. 0.7% of runner-ups).
- **Event/censoring bias:** SMD = 0.17 (duration),  $-0.15$  (start year) between event and censored groups — covariates reasonably balanced (SMD  $< 0.20$ ); the dominant bias is structural right-censoring of 2022–2024 contracts.
- **Manual validation (executed 2026-07-02):** 150 stratified cases audited against the full official BOAMP records (`event_validation/outputs/manual_validation_audit_labeled.csv`). Result: TP 14 / FP 82 / uncertain 4 on 100 linked pairs (precision  $\approx 0.15$  raw,  $\approx 0.09$  weighted to the 665-event population; 0.50 in the HIGH/strict tier; 1.00 at text similarity  $\geq 0.80$ ); 6 of 47 decided unlinked sources hid a plausible missed renewal (12.8%). This is a sample-based audit estimate, not population ground truth; it invalidates the LOW tier and weakens the MEDIUM tier as renewal evidence.

### 5.2 Key limitations

- **Proxy event.** `event = 1` means a plausible renewal link was found, not a legally certified renewal. By composite score, 39.8% of the 665 links are below 0.50 (low), 45.3% between 0.50 and 0.70 (medium), and 14.9% at or above 0.70 (high).
- **Residual buyer fragmentation.** SIREN enrichment consolidated 724 of the 1,210 eligible rows onto a SIREN key (plus 2 SIRET), but 484 remain name-keyed, so spelling variants not resolved by the API can still create false negatives. Enrichment is not ground truth.
- **Short-text false positives.** Generic descriptions (“prestations informatiques”) can match spuriously.
- **Structural right-censoring.** Contracts from 2021–2024 are largely censored; the positive `start_year` hazard partly reflects study truncation, not a genuine cohort trend.
- **Modest discrimination.** Concordance  $\approx 0.65$  — useful for prioritisation, not sufficient for high-stakes individual decisions.

## 6 Completed Work and Remaining Work

**Completed.** BOAMP acquisition and cleaning; buyer SIREN enrichment (2026-06-24); renewal-event reconstruction (665 events, 55.0%); survival modeling (KM, Cox, AFT) with sensitivity and robustness checks; operational 12/24-month risk indicators; rule-based technology taxonomy.

**Remaining.**



- **Act on the manual-validation findings.** The 150-case audit (executed 2026-07-02) estimates linking precision at only  $\approx 9\text{--}15\%$ . The priority is now to re-link with a higher text floor ( $\geq 0.5\text{--}0.65$ ), stop crediting generic CPV codes as exact matches, de-anchor the temporal window from ceiling durations, and re-estimate the survival models on the corrected events.
- **Trained NLP classifier.** The technology taxonomy is rule-based (CPV + keywords); the planned learned classifier was not implemented.
- **Change-point / trend detection (Phase 4).** Scoped but not started.
- **DECP semantic linking.** Applied to BOAMP only; DECP linking remains at the lexical prototype stage.
- **Live prediction scoring.** The current model evaluates contract age as of the study end date; a production run would re-evaluate each contract at a live prediction date.

## 7 Conclusion and Next Steps

The internship delivered a reproducible end-to-end pipeline: BOAMP notice acquisition  $\rightarrow$  cleaning  $\rightarrow$  buyer SIREN enrichment  $\rightarrow$  renewal-event reconstruction  $\rightarrow$  survival modeling  $\rightarrow$  operational risk indicators. The main methodological result is lifting the renewal linking rate from 7.6% (lexical baseline at the same  $W = 6$  window) to 55.0% using Sentence-Transformer embeddings. Survival modeling shows renewal timing is driven by declared contract duration (median 50.1 months), with a marginal category effect, and the log-normal AFT model translates this into actionable 12/24-month renewal-risk rankings.

### Immediate next steps (prioritised):

1. **Re-link with stricter rules and re-model.** The executed manual audit (precision  $\approx 9\text{--}15\%$ ) shows the baseline event definition is too permissive: raise the text-similarity floor, fix generic-CPV scoring, widen the temporal window, then re-estimate KM/Cox/AFT.
2. **Re-score contracts at a live prediction date** (current scores use end-of-study ages).
3. **Review medium/low SIREN matches** to extend buyer consolidation beyond the 724 HIGH-confidence rows already applied.
4. **Replace the rule-based taxonomy** with a trained text classifier.
5. **Implement change-point detection** (Phase 4).